# MCP-Geo Stakeholder Mapping and Value Propositions

## Executive summary

MCP-Geo is a research **Model Context Protocol (MCP)** server that connects MCP-capable AI clients to live **UK geospatial and statistical data**—notably from Ordnance Survey[1] and Office for National Statistics[2]—via a discoverable catalogue of tools with typed schemas, consistent error handling, and observability features (correlation IDs, structured logs, optional Prometheus-style metrics). [3]

The repository is explicit that it is **a personal development / learning project**, **not production code**, and **not approved** by Warwickshire County Council[4] or Department for Science, Innovation and Technology[5]. This strongly shapes positioning: the highest-value stance today is "reference implementation / prototype accelerator" rather than "ready-to-procure service". [6]

From a stakeholder/value perspective, MCP-Geo's distinctive assets are: a broad **tool surface** spanning address/place lookup, administrative boundary lookup (optionally backed by PostGIS), map rendering and vector-tile descriptors, ONS dataset discovery/selection, and data export handoff patterns; a documented **cross-host map delivery baseline** (static first, then overlays, then UI widgets when supported); and an evaluation + trace discipline (question suite, rubric, harness, traffic tracing, UI-event logs). [7]

Assumptions used in this report: the primary intended geography is the United Kingdom[8], inferred from the data sources (OS/ONS/Nomis) and the repository's stated aim of "UK geographic and statistics awareness". No budget, delivery timeline, or deployment region is assumed beyond this. [9]

## Evidence-based repository review

### What MCP-Geo does

MCP-Geo exposes MCP endpoints over both **STDIO (local adapters)** and **HTTP** (a streamable `/mcp` JSON-RPC endpoint), plus dedicated REST-ish convenience endpoints for tools and resources (for example `/tools/list`, `/tools/describe`, `/tools/call`, and `/resources/*`). [10]

The tool catalogue published in the README includes (non-exhaustively):
- Address and UPRN lookup (`os_places.*`) and place-name lookup (`os_names.*`)
- NGD feature queries (`os_features.query`) and linked identifier resolution (`os_linked_ids.get`)
- Map delivery (`os_maps.render`, `os_vector_tiles.descriptor`) plus offline handoff (`os_offline.*`)
- Administrative boundary operations (`admin_lookup.*`)
- ONS observation queries and filtering (`ons_data.*`), dataset search (`ons_search.query`) and explainable selection/ranking (`ons_select.search`)
- Labour-market/statistical access through "Nomis" tool families (`nomis.*`)
- A routing/intention tool (`os_mcp.route_query`) to recommend tools and workflows
- MCP-Apps UI renderers (`os_apps.render_*`) with `ui://` resources for interactive surfaces (geography selector, boundary explorer, route planner, feature inspector, statistics dashboard). [11]

The repo repeatedly frames this as **more than raw API passthrough**, with documented examples of "place profiling" and producing decision-support artefacts (including static and interactive map outputs) as evaluation material. [12]

## Architecture, runtime dependencies, and deployment

The spec package describes a straightforward MCP server architecture: MCP client → MCP-Geo (FastAPI + MCP routers/adapters) → tools layer → upstream OS/ONS APIs and local caches, with logs, correlation IDs, and optional Prometheus-style `/metrics`. [13]

Core dependencies declared include **Python ≥3.11**, **FastAPI**, **Uvicorn**, **Pydantic**, request clients (`requests`, `httpx`), plus geospatial and export helpers (`pyproj`, `pillow`, `openpyxl`). Optional extras add GeoPandas/Shapely and `psycopg` for the boundary cache pipeline. [14]

Deployment paths include:
- Docker container execution (default CMD runs the STDIO adapter) [15]
- Local dev with Uvicorn (`uvicorn server.main:app`) [16]
- MCP client configuration templates via `mcp.json` (stdio, docker, and http registrations). [17]

A notable design choice is the "**compatibility-first**" map-delivery sequence: start with `os_maps.render` (static), then provide deterministic overlay payloads, then use MCP-Apps widgets only if the client advertises UI capability, otherwise degrade through stable fallback skeletons (`map_card`, `overlay_bundle`, `export_handoff`). [18]

## Licensing posture and maturity signals

The README states the project is under an **MIT licence**. [19]
However, the repository root file listing visible on GitHub[20] does not show a standalone `LICENSE` file at the top level (as of 2026-03-08). This is an addressable packaging risk for stakeholder confidence and re-use clarity. [21]

Maturity indicators present in-repo include: semantic versioning (`0.6.0`), structured changelog entries dated **2026-03-08**, and a large body of documentation, evaluation harnesses, and trace tooling; the repository also records a focused public-release secret scan and states "no confirmed secrets found" in a pre-release review. [22]

At the platform level the repo shows **338 commits**, **0 issues**, and **2 open pull requests** at the time of review. [23]

Known maturity gaps are explicitly documented: the completion plan lists the need for CI, stronger client compatibility validation for MCP-Apps, and re-establishing strict quality gates; the testing/quality notes also record periods of static-analysis debt and a time-sliced coverage gate that has not always met the ≥90% target. [24]

## External foundations MCP-Geo depends on

MCP-Geo's value is inseparable from its upstream ecosystems:

- MCP is defined as an open protocol for connecting LLM applications to tools/data sources; MCP-Geo pegs conformance to a specific MCP core revision (`2025-11-25`) and tracks MCP-Apps as a preview extension. [25]
- The OS API surface it uses includes address and place-name products (OS Places/Names), mapping APIs, and an OGC API–Features aligned NGD features service. [26]
- ONS dataset access follows a dataset→edition→version→dimensions→options→observations structure, supporting programmatic discovery and retrieval; MCP-Geo's own research pack specifically leans into this structure for explainable dataset selection. [27]

- Nomis provides a RESTful API supporting SDMX and multiple output formats, which aligns with MCP-Geo's tool abstraction approach. [28]
- The ONS Open Geography portal is framed as the definitive source of geography products and is available under Open Government Licence terms (as described in UK government catalogue material). [29]

## Stakeholder landscape and prioritisation

### Policy and market context shaping stakeholder expectations

The UK's geospatial strategies emphasise cross-sector value (public services, environmental recovery, innovation) and highlight the need to connect people/systems and reduce barriers to combining location data. That aligns directly with MCP-Geo's "uniform access layer" concept. [30]

Separately, government resilience policy explicitly calls out the importance of **mapping risks and systemic interdependencies**, which is a consistent pull for tools that can fuse boundaries, assets, and statistics into decision support. [31]

### Stakeholder types covered and missing types

Covered (as requested): government departments; local authorities; national mapping agencies; emergency services; utilities; transport; environmental NGOs; academia; private sector geospatial firms; consultants; open-source communities; funders.

Material "missing" stakeholder types that typically decide success/failure in real deployments (unspecified in the prompt) include:
- AI-client/platform vendors and integrators (for MCP client compatibility, security posture, UX constraints): OpenAI[32], Anthropic[33], Microsoft[34]. [35]
- Security, assurance, and incident response authorities (prompt-injection and tool-use risk): National Cyber Security Centre[36]. [37]
- Data ethics, citizen trust, and consent/governance owners (especially for location data): Government Digital Service[38] and Geospatial Commission[39]. [40]
- Data protection regulators and legal/commercial licensing owners: Information Commissioner's Office[41].
- Standards bodies and interoperability stewards (in addition to MCP): Open Geospatial Consortium[42] (given OS NGD's OGC API–Features alignment). [43]
- Mapping/tile infrastructure providers and policy bodies when using shared tile services: OpenStreetMap Foundation[44]. [45]

### Stakeholder map table

Scoring method used for prioritisation: Influence (H=3, M=2, L=1) + Interest (H=3, M=2, L=1) + Adoption likelihood (H=3, M=2, L=1). Max score=9.

| Stakeholder group | Influence | Interest | Adoption likelihood | Priority score | Primary "MCP-Geo hook" |
|---|---|---|---|---|---|
| Central government departments & cross-government delivery teams | H | H | M | 8 | Fast, standardised geospatial/statistical tool access for AI-assisted analysis; audit-friendly contracts |

| Stakeholder group | Influence | Interest | Adoption likelihood | Priority score | Primary “MCP-Geo hook” |
|---|---|---|---|---|---|
| Local authorities | M | H | M | 7 | Operational geography lookup + boundary cache + map outputs; reduces specialist friction |
| National mapping & statistics agencies | H | H | M | 8 | Reference integration pattern that increases safe/consistent consumption of official APIs |
| Emergency services & resilience partnerships | H | H | M | 8 | Rapid address → boundary context → map delivery; supports incident triage and situational awareness |
| Utilities & critical infrastructure operators | H | M | L | 6 | Potential to use UPRN/USRN linking + boundary context as a foundation (but needs domain data integration) |
| Transport authorities & operators | M | M | M | 6 | Faster place/feature discovery and map-based comms; integration template for routing/asset overlays |
| Environmental agencies & NGOs | M | M | M | 6 | Lower-friction place/context/area profiling workflows; requires careful licensing and ethics posture |
| Academia & research institutes | M | H | H | 8 | Reproducible evaluation harness + traceability for geospatial AI/tool-use research |
| Private-sector geospatial firms | H | H | M | 8 | Accelerates MCP enablement and “AI+GIS” prototypes; reusable patterns (schemas, fallbacks, observability) |
| Consultancies & system integrators | M | M | H | 7 | “Blueprint plus code” for piloting MCP in the public sector with evidence and documentation |
| Open-source communities & standards bodies | M | H | M | 7 | Contributions to MCP server patterns for maps, provenance, and fallback UI/data contracts |
| Funders & grant-makers | H | M | M | 7 | Scalable demonstrator for geospatial+AI integration, with measurable evaluation artefacts |

## Influence vs interest chart

```
quadrantChart
  title Influence vs Interest for MCP-Geo Stakeholders
  x-axis Low interest --> High interest
  y-axis Low influence --> High influence
  quadrant-1 High influence / High interest
```

```
quadrant-2 Low influence / High interest
quadrant-3 Low influence / Low interest
quadrant-4 High influence / Low interest

"Central gov" : [0.78, 0.90]
"Nat. mapping/statistics" : [0.72, 0.88]
"Emergency services" : [0.80, 0.85]
"Geo firms" : [0.74, 0.83]
"Local authorities" : [0.82, 0.62]
"Academia & research" : [0.78, 0.42]
"Consultancies & integrators" : [0.62, 0.60]
"Funders" : [0.55, 0.82]
"Transport" : [0.60, 0.70]
"Utilities" : [0.50, 0.78]
"Environmental NGOs" : [0.62, 0.35]
"Open-source" : [0.75, 0.40]
```

## Tailored value propositions by stakeholder group

Evidence note: the "how MCP-Geo addresses pain points" mappings below reference MCP-Geo capabilities documented in the README, spec package, and core server code: tool catalogue (`os_places.*`, `admin_lookup.*`, `ons_data.*`, `os_maps.render`, fallbacks), UI resources (`ui://…`), observability (`/metrics`, correlation IDs), and evaluation/tracing. [46]

### Central government departments

Role/interest: policy design and operational delivery where outcomes vary by place; desire to reduce "time to insight" while keeping provenance and audit trails. This aligns with the UK geospatial strategy's emphasis on cross-economy benefits and connecting systems. [30]

Pain points: fragmented access to location data and statistical releases; high dependency on specialist analysts; inconsistent "what geography did you mean?" interpretation; slow map production for briefings.

How MCP-Geo addresses them: intent routing (`os_mcp.route_query`) to recommend workflows; ONS dataset discovery/selection tools (`ons_search.*`, `ons_select.*`, `ons_data.*`) combined with admin geographies (`admin_lookup.*`); map delivery baseline (`os_maps.render` + overlays + fallbacks) to ship consistent outputs to mixed client fleets. [47]

Value proposition statement: **"Turn policy questions into geography-aware, sourced answers faster—using a standard tool interface that can be audited and reused across teams."**

Adoption barriers: research/prototype status and non-endorsement caveat; lack of production governance elements such as RBAC/ABAC (explicitly out of scope for the prototype); prompt-injection/control risks in tool-using AI systems. [48]

Recommended channels: cross-government data/geospatial practitioner networks; digital and security architecture boards; targeted "showcase briefings" using existing repo evaluation artefacts.

Priority ranking: Influence H | Interest H | Adoption M | Score 8.

## Local authorities

Role/interest: service delivery, planning, and place-based prioritisation; heavy use of administrative codes and boundaries; strong need for accessible tooling beyond GIS specialists. [49]

Pain points: staff time spent finding the "right" boundary, joining stats to local geographies, and producing maps; tooling fragmentation (GIS vs spreadsheets vs dashboards); sensitivity to procurement and support overhead.

How MCP-Geo addresses them: fast address and place lookup (`os_places.*`, `os_names.*`), boundary containment/hierarchy (`admin_lookup.*`), and repeatable map + export handoff patterns (`map_card`, `overlay_bundle`, `export_handoff`). PostGIS boundary cache optionally enables performance and offline/deterministic boundary retrieval. [50]

Value proposition statement: **"Make 'where' questions answerable in plain language—without requiring every user to be a GIS expert—while still emitting GIS-grade artefacts when needed."**

Adoption barriers: OS API key management and product entitlements; local operational capacity to run Docker/PostGIS; assurance needs around AI outputs. [51]

Recommended channels: regional local-government data hubs and GIS networks; small pilots attached to real service pain points (housing, accessibility, emergency planning); supplier-engagement days with demonstrable artefacts.

Priority ranking: Influence M | Interest H | Adoption M | Score 7.

## National mapping and statistical agencies

Role/interest: increase correct and ethical use of foundational data; reduce integration friction; support consistent consumption via modern interoperable interfaces.

Pain points: downstream users struggle with discovery, correct parameters, geography mismatches, and combining datasets; support burden; risk of misuse or misinterpretation of authoritative data.

How MCP-Geo addresses them: provides a single interoperable tool abstraction with schema metadata and consistent error handling; documents map delivery reliability patterns across MCP host capability modes; includes dataset-selection research for explainable selection and comparability warnings. [52]

Value proposition statement: **"A reusable 'AI connector' reference design that makes your APIs easier to consume correctly—while preserving provenance, comparability cues, and stable contracts."**

Adoption barriers: reputational risk if mis-positioned as "official"; licensing constraints; need to align with internal API governance; the repo's explicit non-production caveat. [53]

Recommended channels: developer-relations and API platform teams; joint "reference implementation" discussions; curated demonstrations using official example questions and boundary cache workflows.

Priority ranking: Influence H | Interest H | Adoption M | Score 8.

## Emergency services and resilience ecosystems

Role/interest: rapid and accurate location context for incident triage, dispatch support, and multi-agency situational awareness.

Pain points: time pressure; inconsistent address inputs; need to understand “what area is this in?” quickly; mixed technology estates; security constraints.

How MCP-Geo addresses them: exposes address lookup and nearest queries, boundary containment and hierarchy navigation, and map outputs via a baseline-first delivery path that can degrade to non-interactive but reliable artefacts. The OS Places product itself explicitly frames emergency-services value (control-room identification of closest properties). [54]

Value proposition statement: **“From an incident location to a shared operational picture—fast, consistent, and delivered in formats that work even when the client UI is limited.”**

Adoption barriers: operational security approvals; prompt-injection and “confused deputy” risks when tool-using AI is exposed to untrusted inputs; need for rigorous consent and sandboxing for local MCP servers. [37]

Recommended channels: resilience and civil-contingencies exercises; control-room modernisation programmes; targeted pilots around call triage and area-based escalation thresholds.

Priority ranking: Influence H | Interest H | Adoption M | Score 8.

## Utilities and critical infrastructure operators

Role/interest: asset-heavy operations where location is central (maintenance, outages, risk, planning); high value in joining asset registers with administrative boundaries and population indicators.

Pain points: complex internal geospatial stacks; strict data access controls; many proprietary datasets; limited appetite for “prototype” tooling unless it demonstrably reduces integration costs.

How MCP-Geo addresses them today: provides foundational identifier and boundary context patterns (notably linked identifiers and admin boundary caches) and can export artefacts for downstream GIS tooling; but it does not natively include utility asset datasets, so value is strongest as an integration template, not an out-of-the-box solution. [50]

Value proposition statement: **“A reusable integration pattern for making controlled geospatial services ‘AI-callable’—without rewriting every downstream interface.”**

Adoption barriers: missing domain-specific data connectors; licensing/contractual constraints; security posture requirements.

Recommended channels: innovation labs and joint proofs-of-concept with clear data boundaries; architecture workshops focused on extending the tool catalogue safely.

Priority ranking: Influence H | Interest M | Adoption L | Score 6.

## Transport authorities and operators

Role/interest: network planning, disruption management, and place-based communications.

Pain points: slow collation of place/network context; difficulty producing consistent briefing maps; complex joins between domains (roads/paths/administrative areas/stats).

How MCP-Geo addresses them: fast place lookup; feature queries; map delivery baseline and fallbacks; optional sidecar profile for scaled/self-hosted map delivery. [55]

Value proposition statement: **“Make transport disruption and planning questions geography-aware by default, producing maps and exports that are consistent across teams and tools.”**

Adoption barriers: transport-specific network datasets are not first-class in the current toolset; production governance requirements.

Recommended channels: small pilots anchored in recurring operational questions (disruption areas, accessibility to services, corridor profiling).

Priority ranking: Influence M | Interest M | Adoption M | Score 6.

## Environmental agencies and NGOs

Role/interest: place-based environmental outcomes, public engagement, and evidence-led advocacy.

Pain points: varied geospatial capability; frequent need for boundary-aligned indicators; sensitivity to public trust and ethics; licensing constraints for some basemaps/data.

How MCP-Geo addresses them: structured, discoverable tools; ability to output maps and narrative evidence; possibility to combine ONS/Nomis indicators with selected geographies. Public confidence guidance (location data ethics) underscores the need for transparency, accountability, and fairness when using location data. [56]

Value proposition statement: **“Lower the barrier to trustworthy, place-based environmental evidence—while keeping outputs explicit about provenance and limits.”**

Adoption barriers: dependence on OS products that may require entitlements; governance and ethics requirements; reputational risk if maps are over-claimed.

Recommended channels: partnerships with academic labs; open-data-focused pilots; communications-led demos showing explicit limits and provenance.

Priority ranking: Influence M | Interest M | Adoption M | Score 6.

## Academia and research institutes

Role/interest: reproducible research on geospatial reasoning, tool-using agents, and decision-support UX; interest in evaluation harnesses, traceability, and open artefacts.

Pain points: lack of standardised, reproducible interfaces to real data; difficulty comparing agent behaviours across hosts; evaluation data and run artefacts often not preserved.

How MCP-Geo addresses them: an explicit evaluation framework (question suite + rubric + harness), trace proxies for STDIO/HTTP, and documentation packages targeting multiple audiences. [57]

Value proposition statement: **“A reproducible testbed for geospatial tool-use research: standardised tool contracts, measurable evaluation, and trace logging.”**

Adoption barriers: some upstream data requires OS keys/entitlements; research governance for location-linked data.

Recommended channels: research collaborations; workshops and shared benchmark scenario packs; contributions focused on evaluation and safety.

Priority ranking: Influence M | Interest H | Adoption H | Score 8.

## Private-sector geospatial firms

Role/interest: build and sell geospatial products/services; accelerate AI-enabled offerings; reduce bespoke integrations.

Pain points: MCP adoption requires strong interop rules, tool schemas, and safety controls; mixed-client compatibility; proving value quickly without incurring large bespoke costs.

How MCP-Geo addresses them: provides working patterns for tool schema introspection, search and toolsets, structured errors, fallback map contracts, and operational tracing/metrics—valuable as a reference architecture even when firms re-implement in their own stacks. [58]

Value proposition statement: **"A concrete reference implementation showing how to package geospatial capability as MCP tools—complete with reliability fallbacks and evaluation artefacts."**

Adoption barriers: firms will require commercial support models, clearer licensing packaging, and production-grade security/auth controls; the current repo caveat requires careful framing. [59]

Recommended channels: partner programmes; technical deep-dives focused on map delivery reliability and dataset selection explainability; co-authored case studies.

Priority ranking: Influence H | Interest H | Adoption M | Score 8.

## Consultancies and system integrators

Role/interest: implement data/AI solutions across organisations; need reusable blueprints with evidence.

Pain points: moving from "AI demo" to governed pilot; integration complexity; security scrutiny; need for documentation for non-technical stakeholders.

How MCP-Geo addresses them: unusually strong documentation density (spec package, public-sector-readable documentation set), plus security review evidence and trace strategies that help with governance conversations. [60]

Value proposition statement: **"A pilot-ready blueprint: code + specification + evidence that de-risks early-stage MCP adoption conversations."**

Adoption barriers: production hardening tasks remain (CI, auth, stronger quality gates, multi-tenant controls). [61]

Recommended channels: architecture and assurance working groups; procurement discovery phases; capability demonstrations tied to measurable KPIs.

Priority ranking: Influence M | Interest M | Adoption H | Score 7.

## Open-source communities

Role/interest: build reusable patterns for MCP servers, geospatial tooling, and safe tool-use design.

Pain points: many MCP servers are narrowly scoped; few provide rigor around map delivery, provenance, and mixed UI capability.

How MCP-Geo addresses them: contributes patterns for deterministic fallback map contracts, tool metadata/search/toolsets, resources with ETags, and host trace capture—useful beyond the UK domain. [62]

Value proposition statement: **“Raise the bar for MCP server reliability in map-heavy domains: stable contracts, fallbacks, and observability.”**

Adoption barriers: contributor capacity; need to cleanly separate UK-specific connectors from reusable framework code; clarity on licensing packaging.

Recommended channels: MCP server directories and reference implementation discussions; targeted pull requests around CI, documentation drift fixes, and security improvements.

Priority ranking: Influence M | Interest H | Adoption M | Score 7.

### Funders and grant-makers

Role/interest: enable scalable adoption of high-value geospatial+AI integration; seek measurable outcomes.

Pain points: hard to assess “AI integration” proposals without credible evaluation; risk of funding demo-ware.

How MCP-Geo addresses them: already includes a measurable evaluation harness and documented artefacts, offering a more accountable basis for funding decisions and milestone tracking than many prototypes. [63]

Value proposition statement: **“A demonstrator with built-in measurement and transparency—suitable for structured pilots and learning investment.”**

Adoption barriers: needs a clearer route to production governance and sustainability models.

Recommended channels: innovation funding calls; challenge prizes structured around measurable pilots; co-funded evaluation packs.

Priority ranking: Influence H | Interest M | Adoption M | Score 7. (Example UK funder: UK Research and Innovation[64].)

## Positioning, partnerships and pilots

### Product positioning recommendations

Position MCP-Geo explicitly as a **reference implementation for “AI-to-geospatial” interoperability in the UK**, not as a replacement for GIS platforms. This is consistent with the repository caveat and reduces reputational risk. [65]

A robust positioning ladder, aligned to what is already in the repo: - **Layer one (now):** “Prototype + evidence pack” — tool catalogue, spec package, evaluation and trace artefacts, demonstrator reports. [66]
- **Layer two (near-term):** “Reference architecture” — stable contracts, CI, clearer licensing packaging, and hardened security posture (including client guidance). [67]
- **Layer three (conditional):** “Operational service” — only after adding governance controls explicitly called out as non-goals in the prototype (multi-tenant auth, RBAC/ABAC). [68]

### Partnership pathways

High-leverage partnership directions (ordered by expected impact on real adoption): - Align with national data providers (OS/ONS/Nomis ecosystem) around **reference integration patterns** and reducing user error rates (schemas, error taxonomies, explainability in dataset selection). [69]
- Align with geospatial strategy and policy owners on "safe and ethical use" narratives—explicitly drawing on published guidance about public confidence in location data. [70]
- Align with MCP ecosystem stewards on security and client capability negotiation, because real deployments will be constrained by what clients safely support (particularly for UI resources and local server execution). [71]

### Pilot project recommendations

Pilot selection should favour workflows that MCP-Geo already demonstrates well (address→context expansion, profiling custom areas, stable map delivery) and that can be evaluated with existing harness patterns.

Recommended pilots: - **Local authority "Place profiling brief in a week"**: start with a service question, define an area of interest, generate repeatable maps + exports; measure time-to-brief and stakeholder satisfaction using saved artefacts and trace logs. [72]
- **Resilience / emergency planning scenario pack**: ingest a small boundary set into PostGIS, validate containment/hierarchy queries, generate map cards and overlays for an exercise; explicitly align outcomes to resilience policy emphasis on mapping chronic risks. [73]
- **ONS dataset selection for a real policy question**: test the explainable-ranking approach (`ons_select.*`) against analyst judgement; measure "time-to-correct-dataset" and comparability error rate (metrics already proposed in the research pack). [74]

## Measurement and KPIs

A credible KPI set should combine product reliability, user value, and governance/assurance signals. MCP-Geo already contains the scaffolding for several of these (evaluation harness, tracing, optional Prometheus metrics). [75]

Recommended KPI groups:

Operational reliability KPIs: - Tool-call success rate and error-code distribution (using the structured error taxonomy and correlation IDs). [76]
- Latency distributions (p50/p95) and request volumes via `/metrics` when enabled. [77]
- Cache effectiveness for boundary operations: cache coverage, staleness flags, and "degraded performance" indicators already described in cache status outputs. [78]

User-value KPIs: - Time-to-first-correct-answer for a defined question set (use evaluation harness baseline and pilot-specific question packs). [79]
- Reduction in specialist analyst involvement per "where" query (pilot metric, gathered qualitatively and via trace analysis). [80]
- Map output usability across host profiles: pass/fail rates against the map delivery support matrix and fallbacks. [81]

Governance and assurance KPIs: - Secret scanning pass rate in CI (the repo already performed a pre-release scan and recommends adding automated scanning). [82]
- Audit completeness: proportion of pilot sessions with captured MCP traces + UI event logs (where applicable). [80]

## Risks, constraints and mitigations

### Security and prompt-injection risk

Tool-using AI systems introduce distinct prompt-injection and “confused deputy” risks; the UK’s cyber authority warns that LLMs do not enforce a robust boundary between instructions and data inside prompts, meaning mitigations differ from classic injection classes. [83]
MCP’s own security best practices explicitly warn about risks from local MCP servers and recommend consent mechanisms, sandboxing, and scope minimisation. [84]

Mitigations for positioning and pilots: - Treat MCP-Geo as **local-first** and constrain HTTP deployments behind auth where used.
- Adopt “least privilege” principles: narrowly scoped API keys; avoid mixing high-risk tools in a single environment.
- Bake trace and audit capture into pilots by default (stdio/http trace proxies, UI-event logs). [85]

### Data ethics and public trust risk

UK government guidance on location data ethics emphasises that location data is ubiquitous and ethically sensitive, and frames the need for accountable, transparent approaches to use. [86]
Mitigation is as much “product narrative” as engineering: position MCP-Geo outputs as decision support with explicit caveats, provenance, and limits (which the repo’s showcase reports already attempt to model). [87]

### Licensing and upstream dependency risks

MCP-Geo depends on upstream APIs (OS/ONS/Nomis) and, for some map paths, OpenStreetMap tile infrastructure. The OpenStreetMap tile usage policy states capacity is limited, service is best-effort, and heavy/inappropriate use may be blocked; it also requires clear attribution, valid user agents, and caching. [88]
Mitigations: - For any scaled pilot, avoid default reliance on shared `tile.openstreetmap.org`—use a compliant alternative or self-hosted tiles.
- Make licensing posture explicit per connector (OS vs ONS Open Government Licence sources) and reflect that in tooling documentation. [89]

### Product maturity and sustainability risks

The repo itself documents remaining work to reach production-grade completeness (CI pipeline, stricter quality gates, broader client validation for MCP-Apps). [90]
Mitigations: - Publish a short “product readiness matrix” that distinguishes prototype, reference architecture, and production requirements (and map each backlog item to stakeholder risk reduction).
- Address packaging hygiene early (add an explicit licence file; remove doc drift such as references to an ONS API key that do not appear in current Settings). [91]

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[1] [26] https://www.ordnancesurvey.co.uk/products/os-places-api

https://www.ordnancesurvey.co.uk/products/os-places-api

[2] [30] [32] [70] https://www.gov.uk/government/publications/uk-geospatial-strategy-2030/uk-geospatial-strategy-2030

https://www.gov.uk/government/publications/uk-geospatial-strategy-2030/uk-geospatial-strategy-2030

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